

23. (d) I. Let three natural numbers are
 $m = 16, n = 15, k = 20$

$$\therefore g_{m(nk)} = g_{16(300)} = 4$$

$$\text{and } g_{(mn)k} = g_{240(20)} = 20$$

$$\therefore g_{m(nk)} \neq g_{(mn)k}$$

II. $g_{mn} g_{nk} = g_{16,15} g_{15,20} = 1 \times 5 = 5$
 and $g_{mk} = g_{16,20} = 4$
 $\therefore g_{mn} g_{nk} \neq g_{mk}$

So, neither I nor II is correct.

24. (c) I. Let $a = 4$, and $b = 10$

$$\therefore a + b = 14$$

$$\text{HCF } (4, 10) = 2$$

$$\text{and HCF } (14, 10) = 2$$

$$\therefore \text{HCF } (a, b) = \text{HCF } (a + b, b)$$

II. Let $a = 6$ and $b = 15$

$$\therefore b - a = 15 - 6 = 9$$

$$\text{HCF } (6, 15) = 3$$

$$\text{HCF } (6, 9) = 3$$

$$\therefore \text{HCF } (a, b) = \text{HCF } (a, b - a)$$

25. (a) Second number = $\frac{\text{LCM} \times \text{HCF}}{\text{First number}}$

$$= \frac{72 \times 12}{24} = 36$$

$$\therefore \text{Difference between two numbers} = 36 - 24 = 12$$

26. (b) Let two numbers be $29x$ and $29y$.

$$29x + 29y = 232 \Rightarrow x + y = 8$$

$$\therefore (x, y) = (1, 7), (3, 5)$$

Hence, one such pair is 87 and 145.

and the other pair is 203 and 29.

27. (d) Here, $18 = 2 \times 3 \times 3$ and $15 = 3 \times 5$

$$\text{HCF of } 18 \text{ and } 15 = 3$$

$$\text{LCM of } 18 \text{ and } 15 = 2 \times 3 \times 3 \times 5 = 90$$

$$\therefore \text{Product of HCF and LCM of both numbers} = 3 \times 90 = 270$$

28. (a) Given that, three planets revolves the Sun once in 200, 250 and 300 days, respectively in their own orbits.

$$\therefore \text{Required time} = \text{LCM of } 200, 250 \text{ and } 300 = 3000 \text{ days}$$

Hence, after 3000 days they all come relatively to the same position at a certain point of time in their orbits.

29. (b) Given, LCM of two numbers = 2376

$$\text{HCF of two numbers} = 33$$

$$\text{One of the number} = 297$$

$$\therefore (\text{HCF of two numbers}) \times (\text{LCM of two numbers})$$

$$= (\text{First number}) \times (\text{Second number})$$

$$\therefore \text{Second number} = \frac{33 \times 2376}{297} = 264$$

30. (a) It is given that, the HCF of two numbers is 98. This means that both the numbers are multiples of 98. Therefore, the sum of these two numbers must also be a multiple of 98. Among all the four options given, only option (a) satisfies this condition.

31. (a) For integers a, b and c , if $\text{HCF } (a, b) = 1$ and $\text{HCF } (a, c) = 1$ then, $\text{HCF } (a, b, c) = 1$

32. (a) $\therefore \text{HCF } (a, b) = 5$

$$\text{Let } a = 5x \text{ and } b = 5y$$

$$\therefore 5x + 5y = 65 \Rightarrow x + y = 13$$

$$\therefore \text{Number of pairs of } (x, y) = (1, 12), (2, 11), (3, 10), (4, 9), (5, 8), (6, 7)$$

Hence, number of solutions is less than 65.

33. (b) HCF of $(22n + 7, 33n + 10)$ is always 1.

$$\text{Illustration For } n = 1, \text{HCF } (29, 43) = 1$$

$$\text{For } n = 2, \text{HCF } (51, 76) = 1$$

$$\text{For } n = 3, \text{HCF } (73, 109) = 1$$

34. (c) Time after which they will hit the target together again

$$= \text{LCM of } 5, 6, 7 \text{ and } 8$$

$$= 5 \times 3 \times 7 \times 2 \times 2 \times 2 = 840 \text{ s}$$

Duration after which they will hit target together

$$= \frac{840}{60} = 14 \text{ min.}$$

So, they will hit the target together after 14 min.

Hence, they will hit together again at 9:14 am.

35. (c) $\text{HCF} \left(\frac{a}{\text{HCF}(a, b)}, \frac{b}{\text{HCF}(a, b)} \right)$ is always equal to 1.

Illustration Let the two positive integers be $a = 24$ and $b = 36$.

$$\therefore \text{HCF} \left(\frac{24}{\text{HCF}(24, 36)}, \frac{36}{\text{HCF}(24, 36)} \right)$$

$$\text{HCF} \left(\frac{24}{12}, \frac{36}{12} \right) \Rightarrow \text{HCF}(2, 3) = 1$$

36. (d) LCM of two natural numbers

$$= \frac{\text{Product of } m \text{ and } n}{\text{HCF of } m \text{ and } n} = \frac{552}{24} = 23$$

Here, no set of m and n is possible satisfying the given conditions as LCM is always a multiple of HCF.

37. (c) Given, LCM of two integers is 1237, which is a prime number.

So, their HCF is 1.

38. (c) Given, number of cricket balls

$$= 48 = 2^4 \times 3$$

$$\text{Number of hockey balls} = 72 = 2^3 \times 3^2$$

$$\text{and number of tennis balls}$$

$$= 84 = 2^2 \times 3 \times 7$$

$$\therefore \text{HCF of } 48, 72 \text{ and } 84 = 2^2 \times 3 = 12$$

Now, minimum number of rows

$$= \frac{48}{12} + \frac{72}{12} + \frac{84}{12}$$

$$= 4 + 6 + 7 = 17$$

39. (c) The required numbers are 10, 17, 24, ..., 94.

Total number of numbers is 13.

$\therefore$ Sum of these numbers

$$= \frac{13}{2} [10 + 94] = \frac{13}{2} \times 104$$

$$= 13 \times 52 = 676$$

40. (d) Here, $6 - 5 = 1, 8 - 7 = 1$

$$12 - 11 = 1$$

$$\text{LCM of } 6, 8 \text{ and } 12 = 24$$

Required number

$$= 24k - 1, k \text{ is any natural number}$$

$$\text{For } k = 6, \text{the number} = 144 - 1 = 143$$

which is multiple of 13

$$\text{So, sum of digits} = 1 + 4 + 3 = 8$$

41. (a) Let other number be b and HCF be x .

$$\Rightarrow \text{LCM} = 12x$$

$$\text{We have, } x + 12x = 403 \Rightarrow 13x = 403$$

$$\therefore x = 31$$

$\therefore$ Product of two numbers

$$= \text{LCM} \times \text{HCF}$$

$$\therefore 93 \times b = x \times 12x$$

$$\Rightarrow 93 \times b = 12 \times 31 \times 31$$

$$\therefore b = 124$$

42. (d) a, b and c are natural numbers.

I. $\text{LCM of } (ab, ac) = abc$

$$a \times \text{LCM of } (b, c) = abc$$

Hence, statement I is correct.

II. $\text{HCF}(ab, ac) = a \text{HCF}(b, c)$

$$\text{HCF of } (ab, ac) = \text{Common factor of } (ab, ac)$$

$$\text{and } a \times \text{HCF}(b, c) = a \times \text{common factor of } (b, c)$$

Hence, statement II is correct.

III. We know that HCF is always less than LCM.

Hence, statement III is correct.

IV. $\text{HCF}(a, b)$ divides $\text{LCM}(a, b)$

because a common factor between a, b always divides $(a \times b)$.

Hence, statement IV is correct.

04

DECIMAL FRACTIONS

Regularly (1-2) questions have been asked from this chapter. It is one of the most common chapters which we are studying from the very starting age.



FRACTION

If any unit is divided into some parts, then each of these parts is called fraction of that unit. A fraction is represented as p/q , where $q \neq 0$ and here q is called as denominator and p is called as numerator.

e.g. $1/2$, $3/4$, $6/7$ etc. are fractions.

Simple fraction

The fraction which has denominator other than power of 10, is called simple fraction. e.g. $\frac{6}{5}$, $\frac{2}{9}$ etc.

Note Simple fraction is also known as vulgar fraction.

DECIMAL FRACTION

Fraction that has powers of 10 in the denominator are called decimal fraction. e.g.

- (i) $\frac{1}{10}$ is the tenth part of 1 is written as 0.1. (ii) $\frac{1}{100}$ is the hundredth part of 1 is written as 0.01.

Types of Decimals

1. **Recurring Decimals** A decimal number in which a digit or set of digits repeats regularly is called non-terminating repeating decimals or recurring decimals. To represent these fractions, a line is drawn on repeating digits.

e.g. $\frac{1}{7} = 0.1428571428571 \dots = 0.\overline{142857}$

$$\frac{2}{3} = 0.66666 \dots = 0.\overline{6}$$

$$\frac{4}{3} = 1.33333 \dots = 1.\overline{3}$$

2. **Pure recurring decimal** A decimal fraction in which all the digits after the decimal points are repeated, is called a pure recurring decimal.

e.g. $0.786786786... = 0.\overline{786}$
 $0.77777... = 0.\overline{7}$

EXAMPLE 1. Which one of the following is a non-terminating repeating decimal?

- a. $\frac{13}{8}$ b. $\frac{3}{16}$ c. $\frac{3}{11}$ d. $\frac{137}{25}$

Sol. c. $\because \frac{3}{11} = 0.272727...$

$$= 0.\overline{27}$$

$$\frac{13}{8} = \frac{13 \times 125}{8 \times 125} = \frac{1625}{1000} = 1.625$$

$$\frac{3}{16} = \frac{3 \times 625}{16 \times 625} = \frac{1875}{10000} = 0.1875$$

$$\frac{137}{25} = \frac{137 \times 4}{25 \times 4} = \frac{548}{100} = 5.48$$

From above it is clear that all of these are terminating decimals. Hence, $\frac{3}{11}$ is a non-terminating repeating decimal.

3. **Mixed recurring decimal** A decimal fraction in which some digits after the decimal point are not repeated while some are repeated, is called a mixed recurring decimal.

e.g. $0.179999... = 0.17\overline{9}$
 $0.053939... = 0.05\overline{39}$

Note The decimal expansion of a rational number is either terminating or non-terminating recurring. In other words, a number whose decimal expansion is terminating or non-terminating recurring is a rational number.

Operations on Decimal Fractions, Addition and Subtraction

While making addition or subtraction of decimal fractions the numbers are placed in such a way that the decimal point lie in one column. Then, the numbers can be added or subtracted as usual.

EXAMPLE 2. $7.093 - 3.57 = ?$

- a. 3.523 b. 3.513
c. 3.143 d. 3.532

Sol. a. Here,

$$\begin{array}{r} 7.093 \\ - 3.570 \\ \hline 3.523 \end{array}$$

EXAMPLE 3. $51.3 + 7.078 + 1.38 + 0.9 = ?$

- a. 61.668 b. 59.238
c. 60.658 d. None of these

Sol. c. Here,

$$\begin{array}{r} 51.300 \\ 7.078 \\ 1.380 \\ + 0.900 \\ \hline 60.658 \end{array}$$

Note Adding/Annexing zeroes to the extreme right of a decimal fraction does not change its value.

Multiplication

While multiplying two or more decimal fractions, consider them without decimal point and multiply them as usual. In the product so obtained the decimal point is marked off as many places from the right as in the sum of the decimal places in the given numbers.

EXAMPLE 4. $5 \times 0.25 \times 6.301 \times 0.00394 = ?$

- a. 0.0310324220 b. 0.0310324210
c. 0.0310324250 d. None of these

Sol. c. For the product of $5 \times 0.25 \times 6.301 \times 0.00394$

$$\text{Here, } 5 \times 25 \times 6301 \times 394 = 310324250$$

$$\text{Total decimal places } (2 + 3 + 5) = 10$$

$$\therefore 5 \times 0.25 \times 6.301 \times 0.00394 = 0.031032425$$

Division

While dividing the given decimal fraction by a natural number, divide it without the decimal point to get quotient. Here, in the quotient so obtained, place the decimal point after as many places from the right as are there in the dividend.

For dividing a decimal fraction by a decimal fraction, multiply the dividend and the divisor by a suitable power of 10 to make the divisor a whole number and then proceed as in the previous rule.

EXAMPLE 5. $0.01834 \div 13 = ?$

- a. 141.077 b. 1.41077
c. 0.00141077 d. None of these

Sol. c. Here, for $0.01834 \div 13$

$$\Rightarrow \frac{1834}{13} = 141.077$$

$$\text{So, } 0.01834 \div 13 = 0.00141077$$

EXAMPLE 6. $0.0066 \div 0.22 = ?$

- a. 0.03 b. 0.3 c. 3 d. None of these

Sol. a. $0.0066 \div 0.22$

$$\text{So, } \frac{0.0066 \times 10000}{0.22 \times 10000} = \frac{66}{22 \times 100} = \frac{3}{100} = 0.03$$

To Convert a Pure Recurring Decimal into a Simple Fraction

In order to convert a pure recurring decimal fraction into simple (vulgar) fraction, we write the repeated figures only once in the numerator without decimal point and write as many nines in the denominator as the number of repeating figure.

e.g. $0.\overline{5} = \frac{5}{9}$ or $0.\overline{45} = \frac{45}{99}$

EXAMPLE 7. $1.\overline{27}$ in the form p/q is equal to

- a. $\frac{127}{100}$ b. $\frac{73}{100}$ c. $\frac{14}{11}$ d. $\frac{11}{14}$

Sol. c. $1.\overline{27} = 1 + 0.\overline{27} = 1 + \frac{27}{99} = \frac{99 + 27}{99} = \frac{126}{99} = \frac{14}{11}$

To Convert Mixed Recurring Decimal into a Simple Fraction

The numerator is obtained by taking the difference between the number formed by all the digits after decimal point. (Here, repeated digits are taken only once) and the number formed by non-repeating digits. The denominator is obtained by placing, as many nines as in repeating digits followed by as many zeroes as the number of non-repeating digits.

EXAMPLE 8. The vulgar fraction of $0.12\overline{36}$ is

- a. $\frac{120}{823}$ b. $\frac{102}{825}$ c. $\frac{93}{825}$ d. None of these

Sol. b. $0.12\overline{36} = \frac{1236 - 12}{9900} = \frac{1224}{9900} = \frac{102}{825}$

EXAMPLE 9. What is the value of $1.\overline{34} + 4.1\overline{2}$?

- a. $\frac{133}{90}$ b. $\frac{371}{90}$ c. $5\frac{219}{990}$ d. $5\frac{461}{990}$

Sol. d. $\therefore 1.\overline{34} = \frac{134 - 1}{99} = \frac{133}{99}$ and $4.1\overline{2} = \frac{412 - 41}{90} = \frac{371}{90}$
 $\therefore 1.\overline{34} + 4.1\overline{2} = \frac{133}{99} + \frac{371}{90} = \frac{1330 + 4081}{990} = \frac{5411}{990} = 5\frac{461}{990}$

Comparison of Fractions By Converting in Decimal Form

To compare the fraction using decimal, convert each one of the given fractions in decimal form and now arrange them in ascending or descending order as per the requirement.

By Equating Denominators

For comparison of fractions, take LCM of the denominator of all fractions. So, that their denominators are same. Now, the fractions having largest numerator is the largest fraction.

EXAMPLE 10. Arrange $\frac{2}{3}, \frac{5}{6}, \frac{16}{25}, \frac{3}{7}$ in ascending order.

- a. $\frac{2}{3}, \frac{5}{6}, \frac{16}{25}$ and $\frac{3}{7}$ b. $\frac{16}{25}, \frac{5}{6}, \frac{3}{7}$ and $\frac{2}{3}$
 c. $\frac{3}{7}, \frac{16}{25}, \frac{2}{3}$ and $\frac{5}{6}$ d. None of these

Sol. c. Firstly, we convert each of the fraction in decimal form.

$$\frac{2}{3} = 0.\overline{666}, \frac{5}{6} = 0.8\overline{33}, \frac{16}{25} = 0.64, \frac{3}{7} = 0.\overline{428571}$$

Here, $0.\overline{428571} < 0.64 < 0.\overline{666} < 0.8\overline{33}$

Hence, $\frac{3}{7} < \frac{16}{25} < \frac{2}{3} < \frac{5}{6}$ and so $\frac{3}{7}, \frac{16}{25}, \frac{2}{3}$ and $\frac{5}{6}$ are in ascending order.

EXAMPLE 11. Arrange $\frac{7}{8}, \frac{5}{6}, \frac{6}{7}$ in descending order

- a. $6/7, 5/6, 7/8$ b. $7/8, 5/6, 6/7$
 c. $7/8, 6/7, 5/6$ d. $6/7, 7/8, 5/6$

Sol. c. LCM of 8, 6, 7 = 168

$$\therefore \frac{7}{8} = \frac{7 \times 21}{8 \times 21} = \frac{147}{168} \Rightarrow \frac{5}{6} = \frac{5 \times 28}{6 \times 28} = \frac{140}{168}$$

$$\text{and } \frac{6}{7} = \frac{6 \times 24}{7 \times 24} = \frac{144}{168}$$

$$\therefore \frac{140}{168} < \frac{144}{168} < \frac{147}{168}$$

So, descending order of fractions is $\frac{7}{8}, \frac{6}{7}, \frac{5}{6}$.

HCF and LCM of Decimal Fractions

For finding the HCF or LCM of decimal fractions, first make the given fractions to have same number of decimal places by annexing zeroes, if needed. Now, find the HCF or LCM of the numbers without considering decimal. Finally, in the result mark as many decimal places as are there in each of the given numbers.

EXAMPLE 12. The HCF and LCM of 1.600, 8 and 2.4 is

- a. 0.800 and 24 b. 0.6000 and 38
 c. 0.900 and 42 d. None of these

Sol. a. We can write the given numbers as 1.600, 8.000, 2.400.

So, without decimal, the numbers are 1600, 8000 and 2400.

Now, HCF of 1600, 8000 and 2400 is 800.

$\therefore$ HCF of 1.600, 8.000 and 2.400 is 0.800.

And LCM of 1600, 8000, 2400 is 24000.

$\therefore$ LCM of 1.600, 8.000 and 2.400 is 24.000.

> PRACTICE EXERCISE

1. If $(15.9273 - x) = 11.0049$, then the value of x is
(a) 4.9224 (b) 0.4922 (c) 0.4294 (d) 6.932
2. If $(15.39 + 0.236 + 5.290 + 0.0002) = x$, then the value of x is
(a) 0.20916 (b) 2.0916 (c) 209.16 (d) 20.9162
3. If $175 \times 1.24 = 2.17$, then the value of 1.75×124 is
(a) 217 (b) 0.0217
(c) 2.17 (d) None of these
4. If $111.744 \div 28.8 = 3.88$, then the value of $1117.44 \div 288$ is
(a) 3.88 (b) 0.388 (c) 388.0 (d) 38.8
5. The value of $\frac{(0.5)^4 - (0.4)^4}{(0.5)^2 + (0.4)^2}$ is equal to
(a) 0.9 (b) 0.09 (c) 9 (d) 0.009
6. The value of $\frac{0.004 \times 0.0008}{0.02}$ equals to
(a) 0.000016 (b) 0.00016
(c) 0.0016 (d) None of these
7. Which of the following sets of the fractions is in ascending order?
(a) $\frac{6}{8}, \frac{7}{9}, \frac{5}{6}, \frac{11}{13}$ (b) $\frac{5}{6}, \frac{6}{8}, \frac{7}{9}, \frac{11}{13}$
(c) $\frac{11}{13}, \frac{5}{6}, \frac{7}{9}, \frac{6}{8}$ (d) $\frac{11}{13}, \frac{7}{9}, \frac{6}{8}, \frac{5}{6}$
8. When $0.232323\dots$ is converted into a fraction, then the result is
(a) $\frac{1}{5}$ (b) $\frac{2}{9}$ (c) $\frac{23}{99}$ (d) $\frac{23}{100}$
9. If $3.245 \times 10^k = 0.0003245$, then the value of k is
(a) 4 (b) -4 (c) 3 (d) 5
10. The value of $(0.\overline{6} + 0.\overline{8} + 0.\overline{7})$ is
(a) $2\frac{1}{8}$ (b) $2\frac{1}{9}$ (c) $2\frac{1}{3}$ (d) $1\frac{1}{9}$
11. The value of $(6.\overline{88} - 2.\overline{58})$ is
(a) $4.\overline{30}$ (b) $4.\overline{29}$ (c) $3.\overline{22}$ (d) $4.\overline{38}$
12. The expression $(11.98 \times 11.98 + 11.98 \times x + 0.02 \times 0.02)$ will be a perfect square for x equal to
(a) 0.02 (b) 0.2 (c) 0.04 (d) 0.4
13. If $2.5x = 0.5y$, then the value of $\frac{x+y}{x-y}$ is
(a) -1.3 (b) -1.5 (c) 1.5 (d) 1.3
14. 7.2 exceeds its one-tenth by
(a) 8.48 (b) 5.48 (c) 6.48 (d) 5.28
15. If $2.5252525\dots = \frac{p}{q}$ (in the lowest form), then what is the value of $\frac{q}{p}$?
(a) 0.4 (b) 0.42525 (c) 0.0396 (d) 0.396
16. What decimal of an hour is a second?
(a) 0.25 (b) 0.0256 (c) 0.00027 (d) 0.0125
17. If $\sqrt{5} = 2.24$, then the value of $\frac{3\sqrt{5}}{2\sqrt{5} - 0.48}$ is
(a) 1.68 (b) 16.8 (c) 168 (d) 0.168
18. The value of $\left\{ \frac{(0.1)^2 - (0.01)^2}{0.0001} + 1 \right\}$ is equal to
(a) 1001 (b) 11 (c) 101 (d) 100
19. The value of $\frac{3}{3 + \frac{0.3 - 3.03}{3 \times 0.91}}$ is equal to
(a) 0.75 (b) 1.5 (c) 15 (d) 0.15
20. What should be subtracted from the multiplication of 0.527 and 2.013 to get 1?
(a) 0.939085 (b) 0.060851 (c) 1.91984 (d) 2.16085
21. The value of $0.34\overline{67} + 0.13\overline{33}$ is equal to
(a) 0.48 (b) $0.48\overline{01}$
(c) $0.4\overline{8}$ (d) 0.48
22. The greatest fraction out of $\frac{2}{5}, \frac{5}{6}, \frac{11}{12}$ and $\frac{7}{8}$ is
(a) $\frac{7}{8}$ (b) $\frac{5}{6}$ (c) $\frac{11}{12}$ (d) $\frac{2}{5}$
23. Which of the following fractions is greater than $\frac{3}{4}$ and less than $\frac{5}{6}$.
(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{4}{5}$ (d) $\frac{9}{10}$
24. Consider the following statements:
I. $\frac{1}{22}$ cannot be written as a terminating decimal.
II. $\frac{2}{15}$ can be written as a terminating decimal.
III. $\frac{1}{16}$ can be written as a terminating decimal.
Which of the statement(s) given above is/are correct?
(a) Only I (b) Only II
(c) I and III (d) II and III